ReviewPulse: Aspect-Based Sentiment Analysis

Understanding What Customers Love & Hate About Every Single Product Feature

In Simple Language:
When shoppers buy products online, they write mixed reviews like: *"The battery lasts two days, but the camera is blurry."*

A traditional system averages this into **3 Stars** or calls it **"Neutral"**. That hides the truth: is the battery good? Is the camera good? Nobody knows.

ReviewPulse is an intelligent NLP application that separates product features automatically:
• **Battery:** Positive (+0.36) • **Camera:** Negative (-0.49)
It aggregates all mentions into clear verdicts (**Good, Mixed, Poor**), creates interactive charts, and writes auto-summaries so shoppers do not have to read 500 reviews, and sellers know exactly what feature to fix.

Capability	Traditional Sentiment Analysis	ReviewPulse (Aspect-Based ABSA)
Scope of Analysis	Evaluates entire review as 1 global score.	**Evaluates each feature (battery, screen, price) separately.**
Mixed Sentences	Cancels positive and negative terms out into neutral.	**Splits at contrast words (but, however) to judge parts independently.**
Domain Vocabulary	Standard dictionary misses tech hardware words.	**Custom hardware lexicon for blurry, overheats, sturdy, drains.**
Actionable Insights	Only says 60% happy (no details).	**Pinpoints exact defect: Fix the camera; the screen is great.**

How ReviewPulse Works in 5 Simple Steps:

1. Ingest Upload CSV or paste text. Auto-detects review column.	**2. Clean & Split** Strips HTML/URLs. Splits at but, however, although.	**3. Extract Aspects** spaCy POS tags nouns & maps synonyms (display → screen).	**4. Score Polarity** VADER + domain lexicon calculates score (-1 to +1).	**5. Dashboard** Interactive Plotly charts, verdicts & download in Streamlit.

System Architecture & Data Flow

Multi-Layer Architectural Decomposition of ReviewPulse

[REVIEWPULSE SYSTEM ARCHITECTURE]

1. INPUT LAYER:
- CSV File Upload (with smart encoding & column detection)
- Raw Review Text Area • Built-in Benchmark Datasets (40 / 120 Reviews)

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2. PREPROCESSING LAYER:
- clean_text(): URL Removal + HTML Stripping + Whitespace Normalization
- Sentence Tokenization: spaCy Sentence Boundary Detection (doc.sents)
- Contrast Segmentation: CONTRAST_SPLIT Regex (but, however, although, yet)

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3. CORE NLP ENGINE:
- **Aspect Extraction (spaCy):** POS Tagging (NOUN/PROPN) • Lemmatization • Canonical Synonym Map
- **Sentiment Scoring (VADER):** Clause Polarity (-1 to +1) • Custom Tech Lexicon Injection

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4. AGGREGATION & SYNTHESIS LAYER:
- Grouping per Aspect • Mention Counts • Average Polarity Score • Verdict Engine (Good/Mixed/Poor)
- generate_summary(): Plain-English Pros & Cons Natural Language Generation

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5. PRESENTATION LAYER (STREAMLIT DASHBOARD):
- 4 KPI Metric Cards • Grouped Sentiment Bar Charts • Score Ranking Visualizations
- Interactive Verdict Table • Quote Drill-Down Explorer • Multi-CSV Export

Layer	Module / Function	Role & Technical Details
Data Ingestion	app.py (st.file_uploader)	Reads user CSV with automatic encoding fallback (UTF-8 → Latin-1). Auto-detects columns matching review, text, comment.
Text Normalizer	nlp_engine.py clean_text()	Strips regex patterns rhttp\S+, HTML tags r<.*?>, and trims redundant whitespace.
Clause Splitter	nlp_engine.py CONTRAST_SPLIT	Divides multi-aspect clauses across contrastive markers (but, however, although, though, yet, while).
Aspect Normalizer	nlp_engine.py extract_aspects()	spaCy en_core_web_sm identifies NOUN/PROPN. Filters out generic stop-aspects. Normalizes synonyms (e.g. display → screen, money → price).
Domain Scorer	nlp_engine.py CUSTOM_LEXICON	Augments VADER with hardware words: -2.5 for blurry/drains/laggy, -3.0 for overheats/broken, +2.5 for sturdy/crisp/vibrant.
Verdict Engine	nlp_engine.py verdict()	Calculates sentiment distribution. Assigns Good (≥60% pos), Poor (≥60% neg), or Mixed. Generates natural language summary.

Engineering Deep Dive: How It Is Made

Code-Level Walkthrough of the Key Algorithms in Simple Language

1. The Secret Weapon: Contrast-Aware Clause Splitting

In normal sentiment analysis, a sentence like *"The battery is fantastic but the camera is blurry"* is evaluated as a single unit. The positive word ("fantastic") cancels the negative word ("blurry"), yielding an average score of 0.0 (Neutral).
In ReviewPulse: We use a compiled regex:
CONTRAST_SPLIT = re.compile(r"■(?:but|however|although|though|yet|while|whereas)■", re.I)
This splits the sentence into two independent clauses:
• **Clause A:** *"The battery is fantastic"* → Aspect: **battery** | Polarity: **+0.557 (Positive)**
• **Clause B:** *"the camera is blurry"* → Aspect: **camera** | Polarity: **-0.542 (Negative)**
This is why ReviewPulse delivers genuine feature-level accuracy where standard tools fail.

2. Aspect Extraction & Canonical Synonym Mapping

Consumers use dozens of different words to refer to the exact same hardware component. If an app treated "display", "screen", and "panel" as three separate aspects, the statistics would be fractured.
In ReviewPulse:
1. **POS Filtering:** spaCy identifies tok.pos_ in ("NOUN", "PROPN") and extracts the base lemma.
2. **Stop-Aspect Removal:** Words like *"phone", "product", "day", "time", "thing"* are discarded.
3. **Canonical Mapping:** Synonyms are mapped to one canonical aspect name using a lookup table:
   • "display", "screen" → "screen" • "cost", "price", "money" → "price"
   • "shipping", "delivery" → "delivery" • "cam", "camera", "photo" → "camera"
   • "charging", "charger", "battery" → "battery" • "speed", "performance" → "performance"

3. Sentiment Scoring with Custom Tech Lexicon

Off-the-shelf VADER was trained on social media text. It knows words like "love", "happy", and "sad", but it has **no idea** that "drains", "blurry", or "sluggish" are severely negative for electronics.
In ReviewPulse: We inject custom domain weights into VADER lexicon:
• **Negative Tech Vocabulary:** drains (-2.5), blurry (-2.5), laggy (-2.5), overheats (-3.0), sluggish (-2.5), flimsy (-2.5), unresponsive (-2.5).
• **Positive Tech Vocabulary:** sturdy (+2.5), crisp (+2.5), snappy (+2.5), vibrant (+2.5), smooth (+2.2), durable (+2.5), affordable (+2.5).
This gives ReviewPulse domain intelligence without requiring thousands of dollars in cloud LLM training.

4. Categorical Verdict Assignment Logic

After counting Positive, Negative, and Neutral mentions for each aspect, ReviewPulse applies clear classification criteria:
• **GOOD:** If positive mentions ≥ 60% of total mentions for that aspect.
• **POOR:** If negative mentions ≥ 60% of total mentions for that aspect.
• **MIXED:** If neither positive nor negative reaches 60% (balanced or conflicting feedback).

Benchmark Results & Dashboard Showcase

Real Analysis of 40 Benchmark E-Commerce Reviews

1. Discovered Aspects Summary Table (From sample_reviews.csv)

Aspect	Positive	Negative	Neutral	Mentions	Avg Score	Verdict
screen	8	1	0	9	+0.669	**Good**
price	6	0	1	7	+0.736	**Good**
delivery	4	2	0	6	+0.161	**Good**
performance	3	1	1	5	+0.306	**Good**
sound	3	2	0	5	+0.274	**Good**
design	2	0	0	2	+0.804	**Good**
battery	5	4	0	9	+0.065	**Mixed**
camera	3	3	2	8	-0.047	**Mixed**
build quality	2	2	0	4	+0.084	**Mixed**
customer service	1	2	0	3	-0.004	**Poor**
software	1	2	0	3	-0.119	**Poor**

Auto-Generated Natural Language Summary:
“Users love the design, price, screen. Common complaints are about the software, customer service. Opinions are mixed on battery, camera, build quality.”

2. Interactive Streamlit Dashboard Highlights

• **KPI Cards:** Instant metrics displaying Reviews Analysed (40), Aspects Discovered (13), Overall Positive % (50%), and Avg Sentiment Score (+0.11).
• **Grouped Plotly Charts:** Green (Positive), Red (Negative), and Grey (Neutral) bars comparing volume per feature.
• **Ranked Sentiment Heatmap:** Horizontal bar chart ordered by average sentiment score from highest rated to lowest rated.
• **Quote Drill-Down Explorer:** Pick any aspect (e.g. battery) to see exact positive and negative customer quotes side-by-side.
• **One-Click CSV Export:** Download the full granular results CSV, benchmark dataset (40 reviews), or large multi-product dataset (120 reviews).

Presentation & Viva Defense Guide

Frequently Asked Questions & Model Answers for Examiners

Q1: What is the main difference between ReviewPulse and standard Sentiment Analysis?

Standard sentiment analysis looks at an entire review and classifies it as either positive, negative, or neutral. ReviewPulse performs Aspect-Based Sentiment Analysis (ABSA). It identifies individual product components (screen, battery, camera) and computes localized polarity for each feature, allowing shoppers and businesses to know exactly which features excel and which fail.

Q2: Why is sentence splitting at contrast words (but, however) so crucial?

Customer reviews frequently contain conflicting sentiments in a single sentence (e.g., “Great screen but camera is disappointing”). If you do not split at contrast markers, standard lexical models average the positive and negative scores together, yielding an artificial neutral rating. Splitting separates the praise from the complaint, accurately rewarding the screen and penalizing the camera.

Q3: Why did you use spaCy and VADER instead of an external LLM like GPT-4?

1. **Zero Cost:** Runs completely free on local hardware without paying cloud API fees.
2. **Speed & Latency:** Analyzes hundreds of reviews in milliseconds without network bottlenecks.
3. **Data Privacy:** Customer reviews remain confidential on local machines without uploading to third-party cloud servers.
4. **Determinism:** VADER and spaCy give consistent, reproducible outputs with no hallucinations.

Q4: What are the current limitations of the system?

1. **Complex Sarcasm:** Sarcastic phrases like “Great, it broke after two hours” can mislead lexical analyzers.
2. **Implicit Aspects:** Reviews like “It is too heavy to carry” refer to weight without explicitly mentioning the noun “weight”.

Q5: How can this project be enhanced in the future?

1. Replace or ensemble VADER with a fine-tuned Transformer model (e.g., cardiffnlp/twitter-roberta-base-sentiment-latest).
2. Use dependency parse trees (subject-verb-object) for zero-shot aspect-opinion linking.
3. Deploy containerized on AWS (EC2/ECS + S3) or Streamlit Community Cloud for public production access.